

SOP for Resolving Major Alarm ALARM_7328 on Nokia Provider Edge Router in Delhi-NCR

Node: Nokia-PE-Delhi-NCR-21 | **Vendor:** Nokia | **Severity:** Major | **Alarm Code:** ALARM_7328

Description

This SOP outlines the steps to diagnose and resolve a major alarm, ALARM_7328, on the Nokia Provider Edge Router, Node ID: Nokia-PE-Delhi-NCR-21.

Potential Root Causes

Cause	Probability
High CPU utilization causing routing protocol instability	High
Misconfigured routing policies causing packet loss	Medium
Hardware issues with the router's Line Card or Interface	Low

Diagnostic Commands

```
show system resources cpu -- to check CPU utilization
show routing protocols isis -- to verify ISIS protocol configuration
show routing policies -- to check for misconfigured routing policies
show interface -- to verify interface configuration and statistics
```

Resolution Steps

- Step 1: {'Step': 'Step 1: Check CPU utilization and adjust as needed', 'Action': 'Use the command `show system resources cpu` to check CPU utilization. If high, adjust the CPU management parameters or restart the router.', 'CLI_Command': 'cpu-management parameters adjust --cpu-threshold 80'}
- Step 2: {'Step': 'Step 2: Verify and adjust ISIS protocol configuration', 'Action': 'Use the command `show routing protocols isis` to verify ISIS protocol configuration. If misconfigured, adjust the configuration.', 'CLI_Command': 'isis configuration adjust -- metric 10'}
- Step 3: {'Step': 'Step 3: Check and adjust routing policies', 'Action': 'Use the command `show routing policies` to check for misconfigured routing policies. If misconfigured, adjust the configuration.', 'CLI_Command': 'routing policy adjust -- policy-name policy-1'}
- Step 4: {'Step': 'Step 4: Check and replace the Line Card or Interface if necessary', 'Action': 'Use the command `show interface` to verify interface configuration and statistics. If faulty, replace the Line Card or Interface.', 'CLI_Command': 'interface replace -- slot 1 -- subslot 0/0'}

Post-Check Verification

```
show system resources cpu -- to verify CPU utilization
show routing protocols isis -- to verify ISIS protocol configuration
show routing policies -- to verify routing policies
show interface -- to verify interface configuration and statistics
```